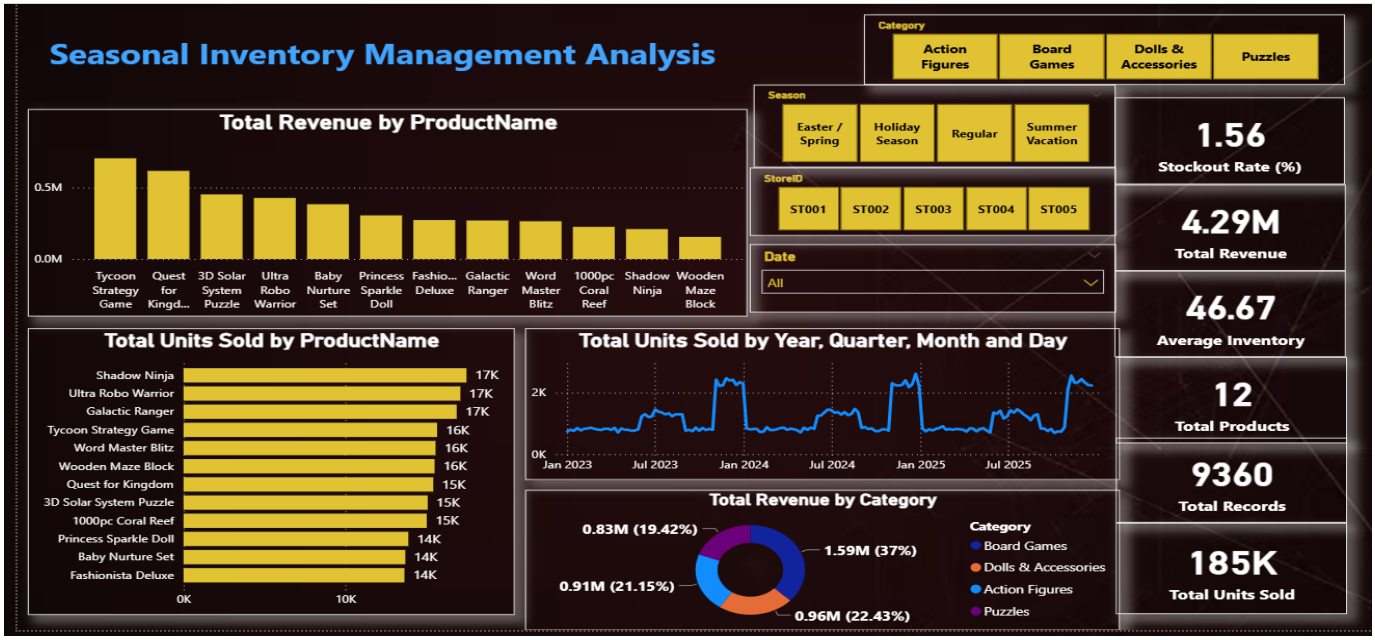
TOYJOY

SEASONAL INVENTORY MANAGEMENT ANALYSIS

Business Case Study Report | Power BI



Prepared by: Junior Analytic Team

Project: Inventory planning, demand seasonality and stock risk
Analysis period visible in dashboard: January 2023 – September 2025

1. Executive Summary

Management overview and headline findings

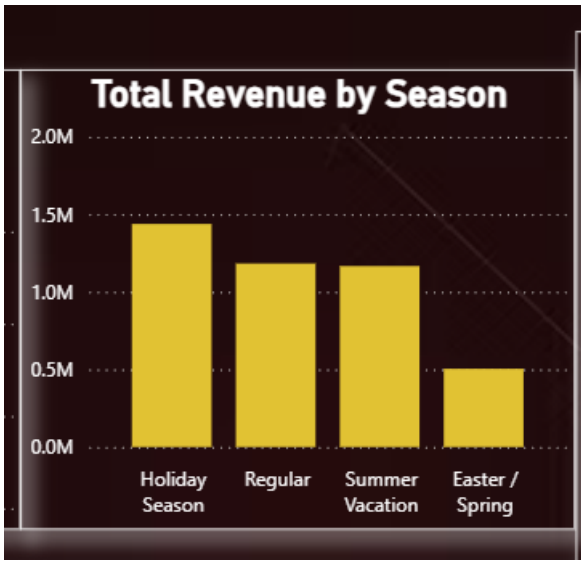
ToyJoy sells toys across multiple stores and product categories. Management needs a clearer view of where inventory is at risk of running out, where capital may be tied up in slow-moving stock, and how demand changes across seasonal periods. This report interprets the completed Power BI dashboard and translates its results into practical inventory actions.

Dashboard KPI	Reported value	Business interpretation
Total revenue	4.29M	Revenue generated across the records in the selected dashboard context.
Total units sold	185,124	Overall sales volume; rounded to 185K on the dashboard.
Average inventory	46.67	Dashboard measure; confirm its unit and aggregation before operational use.
Stockout rate	1.56%	Reported stockout rate; validate numerator/denominator definition.
Total products	12	Distinct product count shown.
Total records	9,360	Number of rows/records represented in the dashboard.

Key findings from the visuals:

→Board Games contribute the largest category revenue: 1.59M (37% of total revenue). Dolls & Accessories contribute 0.96M (22.43%), Action Figures 0.91M (21.15%), and Puzzles 0.83M (19.42%).
→Holiday Season records the highest seasonal unit volume at 60,301 units, followed by Summer Vacation (52,637), Regular (50,545), and Easter / Spring (21,641).
→The product-level table reports every listed product as "Stockout Risk", with weeks of supply between 0.37 and 0.61. This points to a potential replenishment-coverage issue that should be investigated.
→The stockout-observation chart shows the highest counts for 3D Solar System Puzzle and Tycoon Strategy Game, while Fashionista Deluxe and Baby Nurture Set are among the lowest shown.

Overall conclusion: ToyJoy should move toward demand-led replenishment, increase seasonal readiness before the Holiday and Summer Vacation peaks, and validate the stockout and excess-inventory calculations before committing to large inventory transfers or purchases.



ProductName	Easter / Spring	Holiday Season	Regular	Summer Vacation	**Total**
1000pc Coral Reef	1795	3630	4235	5376	**15036**
3D Solar System Puzzle	1818	4012	4239	5027	**15096**
Baby Nurture Set	1704	4252	4241	3516	**13713**
Fashionista Deluxe	1831	3927	4224	3675	**13657**
Galactic Ranger	1814	5676	4061	5336	**16887**
Princess Sparkle Doll	1894	4202	4168	3635	**13899**
Quest for Kingdom	1831	5762	4292	3556	**15441**
Shadow Ninja	1902	6119	4240	5234	**17495**
Tycoon Strategy Game	1798	6193	4232	3456	**15679**
Ultra Robo Warrior	1715	6114	4300	4988	**17117**
Wooden Maze Block	1731	4254	4170	5368	**15523**
Word Master Blitz	1808	6160	4143	3470	**15581**
Total	**21641**	**60301**	**50545**	**52637**	**185124**

2. Business Problem, Objectives and Dataset

Business context

Following the holiday period, management observed both excess inventory and stock availability problems. These conditions create opposing costs: excess stock occupies warehouse space and ties up working capital, while stockouts can cause lost sales, customer dissatisfaction and missed seasonal demand.

Project objectives

1. Identify products and categories that may have excess inventory or weak sales velocity.
2. Understand demand changes by season, month and year.
3. Highlight products with stockout observations or low weeks of supply.
4. Recommend inventory actions that balance service levels, warehouse capacity and cash flow.

Dataset profile

The Power BI model shown in the screenshots uses a table named toyjoy_inventory. The dashboard reports 9,360 records and 12 products, with five store identifiers (ST001–ST005) available as a slicer. The date trend visual spans approximately January 2023 through September 2025.

Visible fields include Category, Date, ProductID, ProductName, Season, StoreID and measures/columns relating to inventory, units sold, weekly sales, weeks of supply, inventory status, excess inventory, revenue, cost of goods sold and stockout observations.

Data preparation considerations

1. Standardize category and season labels so that spelling or spacing variations do not split the same group.
2. Check date data types and create a continuous calendar table for year, quarter, month and day analysis.
3. Inspect missing or negative stock, sales and cost values; document whether missing inventory is imputed, excluded or treated as unknown.
4. Confirm the grain of each row (for example, product-store-date) to avoid double-counting inventory or sales.
5. Validate product and store keys, remove unintended duplicates, and reconcile totals against source records.

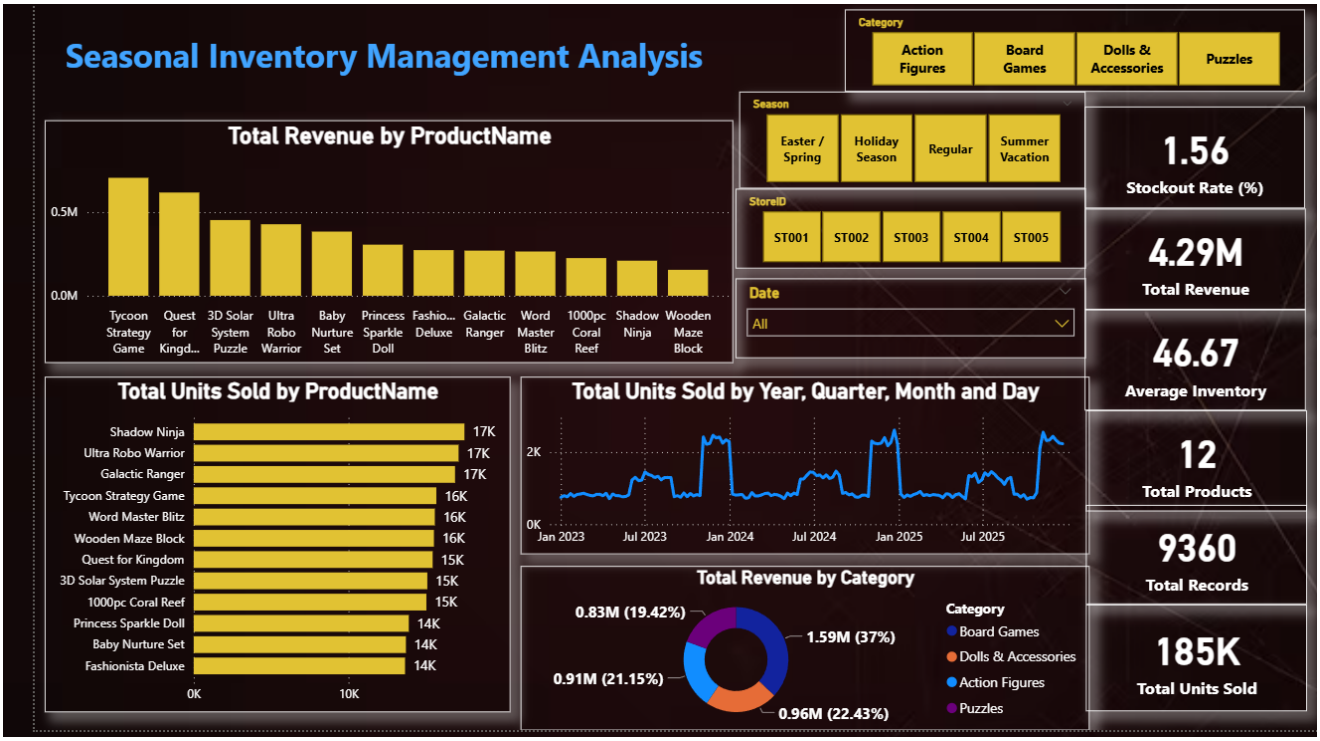
3. Analytical Methodology and KPI Definitions

The report follows a descriptive analytics approach: summarize performance, compare products and categories, inspect stock-risk indicators, and identify recurring seasonal patterns. The Power BI report uses slicers for Category, Season, StoreID and Date, enabling the user to narrow the analysis to a specific business context.

KPI	Typical calculation / interpretation	Decision supported
Units sold	SUM of units sold over the selected period	Identify volume leaders and slow movers.
Revenue contribution	Product/category revenue ÷ total revenue	Prioritize commercially important products.
Inventory turnover	Units sold ÷ average inventory (use consistent time units)	Assess how quickly stock is converted into sales.
Weeks of supply	Available inventory ÷ average weekly unit sales	Estimate how long current stock may last.
Stockout rate	Stockout events or stockout periods ÷ eligible observations	Measure availability problems; define denominator explicitly.
Excess inventory value	Excess units × unit cost	Estimate cash tied up in stock above target.
Seasonal demand	Units sold/revenue grouped by season and time period	Plan inventory ahead of predictable demand peaks.

4. Dashboard Overview and Revenue Performance

The dashboard provides an executive overview. It combines revenue by product, units sold by product, a time-series view, category revenue share, slicers and headline KPI cards.



Revenue by category

Category	Revenue	Share
Board Games	1.59M	37.00%
Dolls & Accessories	0.96M	22.43%
Action Figures	0.91M	21.15%
Puzzles	0.83M	19.42%

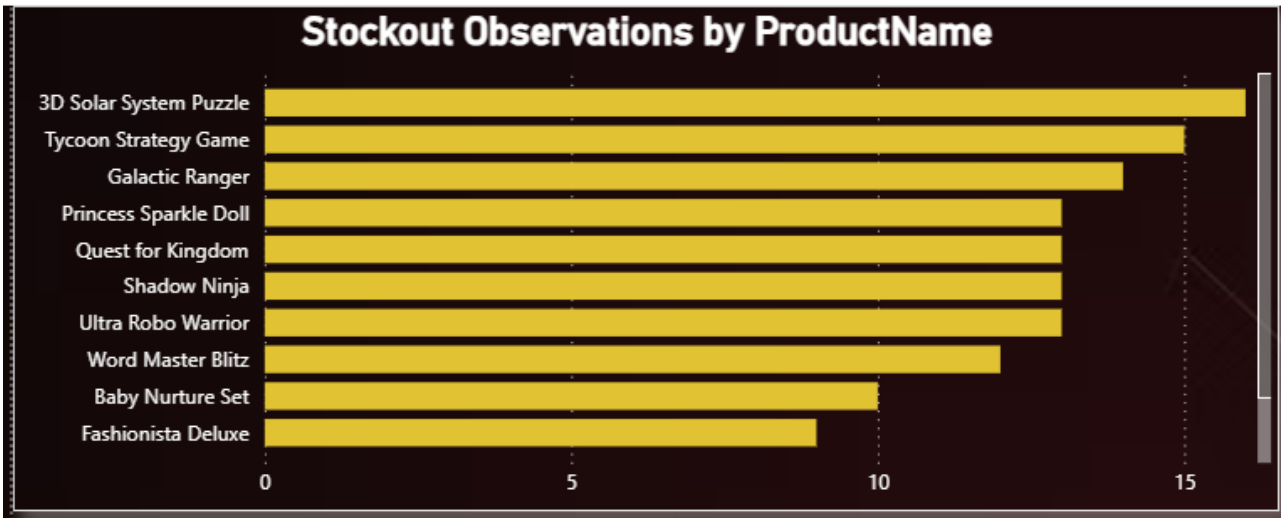
Board Games are the largest revenue contributor, accounting for a little over one-third of total revenue. The remaining three categories have relatively balanced contributions, each near one-fifth to one-quarter of revenue. This suggests that category-level inventory policies should

protect the Board Games revenue base while continuing to monitor demand and stock efficiency in the other categories.

The product revenue visual identifies Tycoon Strategy Game as the largest revenue contributor among the displayed products, followed by Quest for Kingdom and 3D Solar System Puzzle. Exact product-level revenue values are not legible in the supplied screenshot, so the report does not assign precise amounts.

5. Product-Level Inventory and Stockout Analysis

The dashboard combines stockout observations, a detailed product inventory table and weeks of supply by product and store. The detailed table is particularly useful because it connects sales velocity with the stock-coverage indicator.

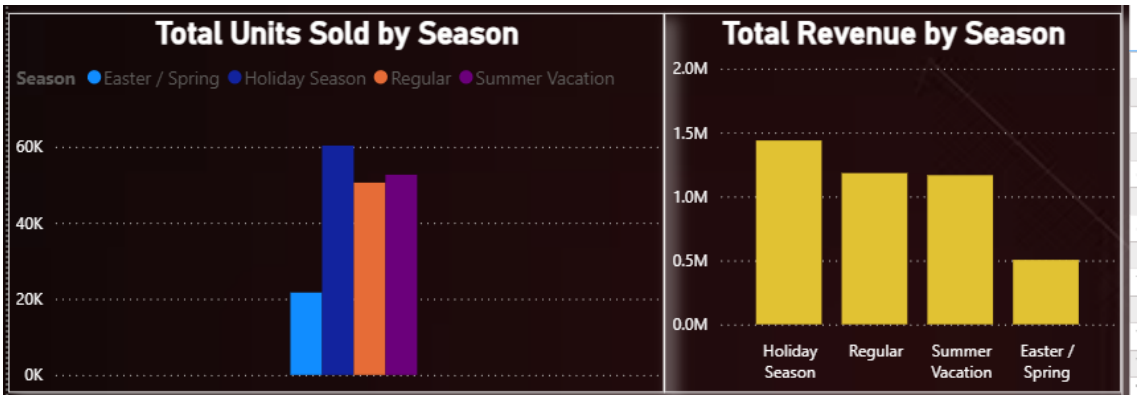


ProductName	Category	Average Inventory	Total Units Sold	Average Weekly Sales	Weeks of Supply	Inventory Status	Excess Inventory Value
Word Master Blitz	Board Games	59.33	15581	99.88	0.59	Stockout Risk	0.00
Wooden Maze Block	Puzzles	59.11	15523	99.51	0.59	Stockout Risk	0.00
1000pc Coral Reef	Puzzles	58.81	15036	96.38	0.61	Stockout Risk	0.00
Shadow Ninja	Action Figures	47.22	17495	112.15	0.42	Stockout Risk	0.00
Galactic Ranger	Action Figures	46.19	16887	108.25	0.43	Stockout Risk	0.00
Tycoon Strategy Game	Board Games	45.01	15679	100.51	0.45	Stockout Risk	0.00
3D Solar System Puzzle	Puzzles	44.02	15096	96.77	0.45	Stockout Risk	0.00
Baby Nurture Set	Dolls & Accessories	42.49	13713	87.90	0.48	Stockout Risk	0.00
Fashionista Deluxe	Dolls & Accessories	41.72	13657	87.54	0.48	Stockout Risk	0.00
Ultra Robo Warrior	Action Figures	40.76	17117	109.72	0.37	Stockout Risk	0.00
Quest for Kingdom	Board Games	38.59	15441	98.98	0.39	Stockout Risk	0.00
Princess Sparkle Doll	Dolls & Accessories	36.69	13899	89.10	0.41	Stockout Risk	0.00
Total		**46.67**	**185124**	**1,186.69**	**0.04**	**Stockout Risk**	**0.00**

Key interpretation:

1. Shadow Ninja and Ultra Robo Warrior have the highest unit sales among the 12 products, and their weekly sales rates are also high. These items warrant close replenishment monitoring.
2. Ultra Robo Warrior has the lowest weeks of supply shown (0.37), followed by Quest for Kingdom (0.39) and Princess Sparkle Doll (0.41). These are the strongest immediate candidates for a stock-coverage review.
3. 1000pc Coral Reef, Wooden Maze Block and Word Master Blitz have the highest weeks of supply in the table (0.61, 0.59 and 0.59), but these values are still below one week. They are relatively better covered, not necessarily overstocked.
4. The stockout-observation chart shows 3D Solar System Puzzle and Tycoon Strategy Game at the top of the displayed list. Investigate whether these observations cluster in particular stores or seasons.

6. Seasonal Demand and Time-Series Analysis



Season	Units sold	Share of total units (approx.)
Holiday Season	60,301	32.6%
Summer Vacation	52,637	28.4%
Regular	50,545	27.3%
Easter / Spring	21,641	11.7%
Total	**185,124**	**100%**

Holiday Season is the strongest demand period, contributing approximately 32.6% of total units sold. Summer Vacation accounts for 28.4%, Regular for 27.3%, and Easter / Spring for 11.7%. The combined Holiday and Summer periods account for about 61% of unit volume, making them the most important seasonal planning windows.

The monthly trend contains visible peaks around December 2023, December 2024 and November 2025 is not shown; the dashboard actually ends around September 2025, with pronounced peaks around December 2023, December 2024 and late 2024/early 2025 depending on the plotted month sequence. The visible line suggests recurring year-end demand spikes and lower-volume intervals between peaks. The monthly chart labels are crowded, so exact month-to-month values should be exported from Power BI for forecasting.

The product-by-season matrix can be used to identify which products drive each seasonal peak. For example, the visible totals show Holiday Season has 60,301 units overall, materially above the other seasons. Use this matrix to create product-specific seasonal uplift factors rather than applying one blanket uplift to every toy.

7. Business Insights and Inventory Recommendations

Immediate actions (next 2–4 weeks)

1. Review the lowest-coverage products first. Validate on-hand, committed, in-transit and safety stock for Ultra Robo Warrior, Quest for Kingdom, Princess Sparkle Doll, Shadow Ninja and Galactic Ranger.
2. Investigate store-level availability. Use the StoreID slicer and weeks-of-supply visual to find stores where high-demand products are under-covered; transfer stock between stores where feasible before placing new orders.
3. Check stockout observations for 3D Solar System Puzzle and Tycoon Strategy Game. Identify the affected dates and stores, then determine whether stockouts were caused by forecast error, delayed replenishment or uneven allocation.
4. Do not liquidate products solely because their unit sales are lower. Compare sales velocity with inventory age, weeks of supply, margin, seasonality and the excess-inventory value measure.

Seasonal planning actions (next 1–2 quarters)

1. Build a pre-season buying calendar for Holiday Season and Summer Vacation. Start supplier and capacity planning before the demand ramp, with a review of lead times and supplier minimum order quantities.
2. Create separate demand forecasts by product and store, using monthly history, season, promotions and trend. Apply seasonal uplift only where historical product-level evidence supports it.
3. Set target service levels by product class: high-volume/high-risk products receive tighter monitoring and higher protection; slower-moving items receive smaller, more frequent replenishment where supplier economics allow.

Long-term inventory control

1. Implement reorder point = expected demand during replenishment lead time + safety stock. Estimate safety stock from demand variability and supplier lead-time variability.
2. Use order-up-to quantity = target stock level − (on-hand + on-order − backorders), subject to warehouse capacity and pack-size constraints.
3. Introduce ABC/XYZ segmentation: ABC by revenue or gross margin contribution; XYZ by demand variability. Combine the segments to prioritize planning effort.
4. Track excess inventory in both units and currency, with ageing buckets and a clear excess threshold. Use markdowns, bundles, channel transfers or supplier returns only after validating margin impact.

Suggested control targets

Establish targets after a baseline period rather than inventing arbitrary thresholds. Recommended controls include weekly coverage review for A-items, monthly forecast-bias review, seasonal readiness checkpoints, stockout root-cause tracking, and a warehouse-capacity ceiling.

8. Conclusion and Assumptions

The dashboard provides a useful starting point for improving ToyJoy's inventory decisions. It shows 185,124 units sold and 4.29M in revenue across 12 products and 9,360 records. Board Games are the largest revenue category, while Holiday Season and Summer Vacation together represent approximately 61% of unit demand. Product-level weeks of supply are low across the displayed list, and stockout observations are concentrated among a subset of products.

The strongest operational message is to protect availability for high-volume products and prepare early for seasonal peaks. However, the supplied screenshots do not establish which products are genuinely overstocked: the visible excess inventory value is zero for all products in the table. Before making stock-reduction decisions, ToyJoy should validate the excess inventory measure and the stockout-rate definition against source transactions.

Assumptions and limitations

1. The dashboard's displayed totals and product table values are treated as the source of truth for this report; values are rounded where shown in the visuals.
2. The visible date trend covers approximately January 2023 to September 2025; confirm exact start/end dates in the source model.
3. The report assumes the seasonal labels represent business-defined demand periods, not necessarily standard calendar quarters.
4. The report does not invent missing product-level revenue, unit costs, lead times, stockout event definitions or forecast accuracy values.
5. Weeks of supply below one week are interpreted as low coverage only if the inventory and weekly sales measures use compatible units and filter contexts.